

Module 04: SQL DML and Integrity Constraints

SQL DML Operations & Relational Integrity Enforcement

Data Manipulation Commands, Constraints Hierarchy & Referential Cascade Rules

1. DML Operations (Data Manipulation)

INSERT INTO (Tuple Insertion)

```
INSERT INTO Student (Roll, Name) VALUES (101, 'Aman');
```

Bulk Insert: `INSERT INTO Backup SELECT * FROM Student;`

UPDATE (Tuple Modification)

```
UPDATE Employee SET Salary = Salary * 1.1 WHERE Dept='IT';
```

Warning: Missing WHERE updates ALL rows in the table!

DELETE FROM (Row Deletion)

```
DELETE FROM Student WHERE CPI < 5.0;
```

Maintains undo log; triggers foreign key cascade checks.

SELECT (Data Retrieval Expression)

```
SELECT Name, CPI FROM Student WHERE Dept='CSE' ORDER BY CPI DESC;
```

Combines relational algebra σ , π , joins, and aggregation into one declarative query.

2. SQL Integrity Constraints & Cascades

The 6 SQL Constraints Matrix:

- NOT NULL: Column cannot store NULL values.
- UNIQUE: All values in column must be distinct (allows NULLs).
- PRIMARY KEY: NOT NULL + UNIQUE. Table identifier.
- FOREIGN KEY: References parent table primary key.
- CHECK: Custom condition validation (e.g.

CHECK (Salary > 0)).

- DEFAULT: Default fallback value when no value is supplied.

Foreign Key Referential Integrity Actions: ON DELETE / ON UPDATE <ACTION>

1. CASCADE: Deleting/updating parent auto-deletes/updates child rows.
2. SET NULL: Deleting parent sets foreign key in child to NULL.
3. SET DEFAULT: Deleting parent resets child FK to its default value.
4. RESTRICT / NO ACTION: Rejects parent deletion if child row exists! Default action in SQL is RESTRICT (prevents orphaned records).

Data Manipulation & Integrity: DML commands operate on data rows, while integrity constraints guarantee ACID database consistency.

1. SQL Integrity Constraints

Constraint	Rule	Enforcement
NOT NULL	Column cannot be blank.	Table column definition.
UNIQUE	No two rows share the same value.	Unique index created.
PRIMARY KEY	NOT NULL + UNIQUE identifier.	Clustered index created.
FOREIGN KEY	Value must exist in parent Primary Key.	Referential integrity check.
CHECK	Validates custom business rule condition.	Predicate validation.

2. Referential Integrity Actions

When a parent key is deleted or updated: CASCADE propagates changes, SET NULL blanks child foreign keys, and RESTRICT rejects parent deletion to prevent orphaned records.